

Metoda celor mai mici pătrate

$\langle f, g \rangle = \text{prod. scalar pt. } f \text{ și } g \text{ (funcții/vectori)}$

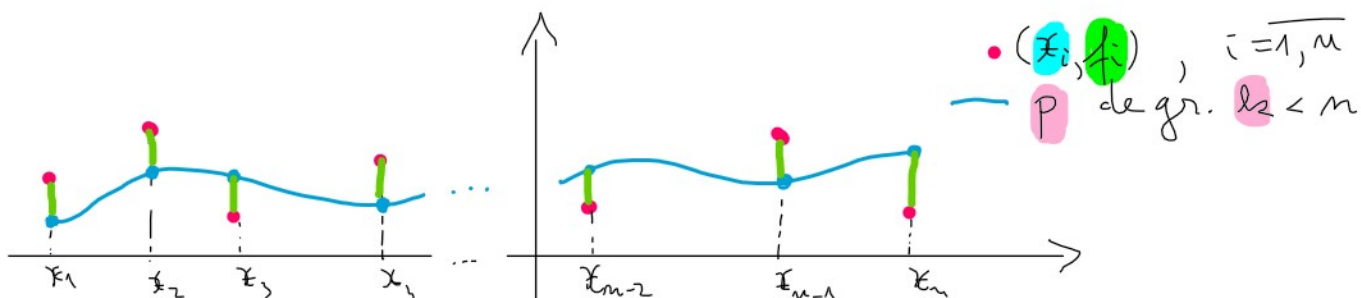
$$\|f - g\| = \text{distanța dintre } f \text{ și } g \text{ (normă)} \\ = \sqrt{\langle f - g, f - g \rangle}$$

Ex:  $\langle f, g \rangle = f_1 \cdot g_1 + \dots + f_n \cdot g_n = \text{prod. scal. Euclidian}$

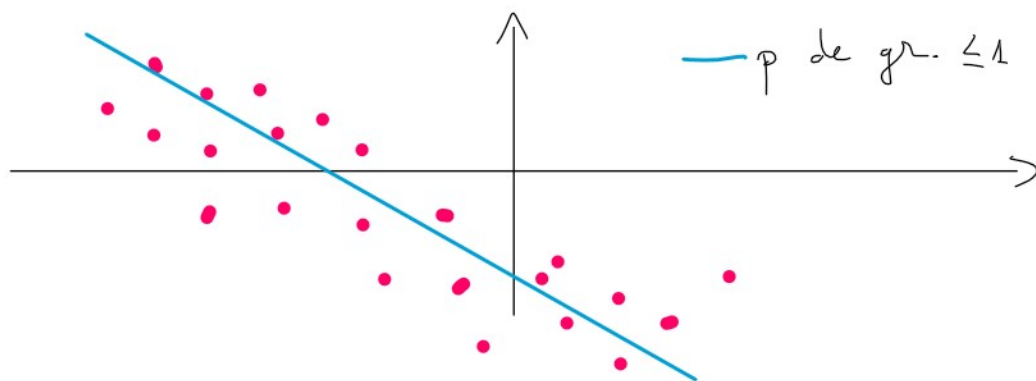
$$\|f - g\| = \sqrt{(f_1 - g_1)^2 + \dots + (f_n - g_n)^2} = \text{norma Euclid.}$$

Ex:  $\langle f, g \rangle = \int_a^b f(x) \cdot g(x) dx$

M. c. m. m. p. discretă (norma Euclidiană)



(?) Găsim  $\hat{p}$  pol. de gr.  $\leq k$  a.î.  $\|f - \hat{p}\| = \min_{\substack{p \in P_k \\ \text{sp. pol.} \\ \text{gr.} \leq k}} \|f - p\|$ .



$$p(X) = c_k \cdot X^k + \dots + c_1 X + c_0 \cdot 1$$

$$\begin{cases} p(x_1) = c_k \cdot x_1^k + \dots + c_1 \cdot x_1 + c_0 \cdot 1 \approx f_1 \\ \vdots \\ p(x_m) = c_k \cdot x_m^k + \dots + c_1 \cdot x_m + c_0 \cdot 1 \approx f_m \end{cases} \quad \left. \begin{array}{l} k+1 \text{ necunosute} \\ \wedge \\ m \text{ ec.} \end{array} \right\}$$

$$\underbrace{\begin{bmatrix} x_1^k & \dots & x_1 & 1 \\ \vdots & & & \vdots \\ x_m^k & \dots & x_m & 1 \end{bmatrix}}_A \cdot \underbrace{\begin{bmatrix} c_k \\ \vdots \\ c_1 \\ c_0 \end{bmatrix}}_c \approx \underbrace{\begin{bmatrix} f_1 \\ \vdots \\ f_m \end{bmatrix}}_f$$

list. supradeterminat:  $A \cdot c \approx f$ ,  $A \in \mathcal{M}_{n,m}(\mathbb{R})$   
 $n \geq m (= k+1)$   
 Gătim  $\hat{c} \in \mathbb{R}^m$  a.î.  $\|A \cdot \hat{c} - f\| = \min_{c \in \mathbb{R}^m} \|A \cdot c - f\|$ .

Descompunerea QR:  $\forall A \in \mathcal{M}_{n,m}(\mathbb{R})$ ,  $n \geq m$ ,

$\exists Q \in \mathcal{M}_{n,m}(\mathbb{R})$ ,  $Q^{-1} = Q^T$ ,  $\exists R \in \mathcal{M}_{n,m}(\mathbb{R})$  triang. superior,

$$R = \begin{bmatrix} \begin{matrix} \times & & \\ 0 & \times & \\ & & \times \end{matrix} & \\ 0 & \end{bmatrix}_{\substack{m \\ m-m}} \quad \text{a.î.} \quad A = Q \cdot R$$

$$\begin{aligned} \bullet \quad Q^{-1} = Q^T &\Rightarrow \|Q \cdot v\| = \|v\|, \forall v \in \mathbb{R}^m \\ \bullet \quad \|A \cdot c - f\| &= \|Q(R \cdot c) - f\| = \|Q(R \cdot c - Q^T \cdot f)\| \end{aligned}$$

$$\|A \cdot c - f\| = \|Q(R \cdot c) - f\| = \|Q(K \cdot c - Q \cdot f)\|$$

$$= \|R \cdot c - Q^T \cdot f\| = \left\| \begin{bmatrix} \hat{y} - \hat{b} \\ -\tilde{b} \end{bmatrix} \right\|$$

$$\begin{bmatrix} R \\ 0 \end{bmatrix} \cdot \begin{bmatrix} c \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} \hat{y} \\ 0 \end{matrix}$$

$$\begin{bmatrix} b_1 \\ \vdots \\ b_m \\ b_{m+1} \\ \vdots \\ b_n \end{bmatrix} \begin{matrix} \hat{b} \\ \tilde{b} \end{matrix}$$

Lös. syst. überdet.

$$A \cdot c \approx f \quad \text{este}$$

$$\hat{y} = \hat{b}$$

$$\Downarrow$$

$$R(1:m, :)\cdot \hat{c} = b(1:m)$$

$$\|A \cdot \hat{c} - f\| = \min_{c \in \mathbb{R}^m} \|A \cdot c - f\| \leftarrow \hat{c} = R(1:m, :) \setminus b(1:m)$$

$$\|\tilde{b}\| = \|b(m+1:n)\|$$